

Maximum-Entropy Effective-Temperature Representatives and Nonlinear Angular Response Ledgers

Anonymous Author¹

¹*Anonymous Institution*

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Finite angular moments select a representative of a massless kinetic state but do not determine every nonlinear observable of the full state. We combine a maximum-entropy inverse-temperature representative with an insertion-resolved angular response ledger. For Maxwell–Boltzmann, Planck Bose–Einstein, and zero-chemical-potential Fermi–Dirac statistics, every interior realizable angular energy-moment vector selects a unique positive representative; statistics alter only a radial constant. Among finite monomial exponent families, only the weight sets $\{1\}$ and $\{0, 1\}$ preserve a finite positive linear Wien tail on a full-dimensional open parameter set. Standard moment-matched entropy-projection identities anchor a coordinate-relative (n, k) ledger that separates representative-generated terms from residual-dependent terms. Shell contraction yields $SO(3)$ -invariant positive-semidefinite Gram ledgers and an exact discarded- L^2 staircase. A positive equal-information family proves the reconstruction limit: identical retained energy moments imply a common representative and representative-generated ledger, but not common number-weighted or higher-radial-weight residual observables. A normalization-controlled expansion shows that the leading residual cost is set by the Fisher norm and that shape dependence first appears at cubic order. A positive energy-matched two-temperature mixture provides a physical anchor: its $p = 4$ energy moment is unchanged pointwise while its $p = 3$ and $p = 5$ moments differ.

I. INTRODUCTION

Moment reduction poses two logically separate questions. The first is representative selection: which distribution should stand for the equivalence class sharing a finite moment vector? The second is observable attribution: after that representative is selected, which terms in a nonlinear observable are fixed by the retained data, and which require information that the moments did not determine? The second question is the focus of this paper. An observed high multipole alone does not answer it, because nonlinear composition of a bandlimited representative can generate arbitrarily fine angular structure order by order.

Maximum entropy provides a canonical answer to the first question and underlies established kinetic closure hierarchies, together with a substantial theory of convex duality, realizability, degeneracy, and numerical conditioning [1–9]. Quantum-statistical and radiative versions are also well developed [10–12], and divergence-based angular closures have been formulated directly on $\mathbb{S}^2$ [13]. These works provide the variational and closure background; we do not propose a new general entropy closure or a closed transport system.

The angular and information-geometric ingredients are likewise inherited. Temperature superpositions and logarithmic temperature moments organize temperature mixing and spectral distortions [14, 15]. Products of spherical harmonics, Clebsch–Gordan selection rules, and rotationally invariant recouplings are standard representation theory [16, 17]. Bregman projections and exponential-family geometry are established structures [2, 18]. The paper-specific contribution is to retain a representative/residual insertion label before angular-shell contraction. That insertion attribution is defined relative to fixed response and residual coordinates and a fixed scaling path; the

downstream Gram contraction, not the (n, k) labeling itself, is invariant under common $SO(3)$ rotations and within-shell orthonormal basis changes. Recent Fisher–Rao work on a relativistic moment manifold uses geodesic distance and curvature to characterize approach to hydrodynamics [19]; our use of Fisher geometry is instead local, conditioned on a selected angular representative, and directed at downstream observable provenance rather than dynamical relaxation.

The construction has four theorem-bearing layers. First, for massless Maxwell–Boltzmann, Planck Bose–Einstein, and zero-chemical-potential Fermi–Dirac statistics, every interior angular energy-moment vector selects a unique positive inverse-temperature representative. The statistics enter only through a positive radial constant, while the natural-parameter cone and its inverse-cubic boundary barrier are common. Second, nested moment sets produce exact entropy-gap and adjacent-resolution identities. Third, analytic observables admit a bivariate amplitude ledger: one index records representative-composition order, one records output multipole, and a residual-insertion count records whether the term is fixed by the representative. Fourth, shellwise Gram contractions yield rotation-invariant positive-semidefinite provenance ledgers and an exact discarded- L^2 staircase.

Most ingredients in those layers are standard when taken separately. The novelty claim is narrower: a moment-selected effective-temperature representative is combined with a data-relative insertion ledger, invariant shellwise contraction, and a positive equal-information family that establishes the reconstruction limit. The ledger answers an attribution question that an ordinary angular spectrum cannot: whether a generated shell is a deterministic nonlinear consequence of the representative or depends on an unidentifiable residual. It does

not estimate that residual, close the dynamics, or provide a time-propagated error bar. Operationally, this distinction prevents a high- L tail generated by the representative from being mistaken for recovered unresolved kinetic information. It identifies which observable sectors are already computable from retained data and which sectors require an independently supplied enrichment.

This use of Gram objects must also be distinguished from recent Gramian moment closures, where Gram matrices of orthogonal polynomials construct the closure relation itself [20]. Here the representative is fixed variationally before any Gram contraction; the Gram matrices are downstream invariant summaries of generated observable terms and do not predict a next primitive moment.

The sharp limitation is demonstrated constructively. We build positive kinetic states with identical retained energy moments and hence the same maximum-entropy representative and the same representative-generated ledger, but distinct residual-sensitive number- and higher-radial-weight fingerprints. Figure 1 summarizes the distinction. The present article is local in spacetime and static in time: it is not a transport closure, a global stability result, or an assertion that higher-order entropy closures do not exist.

II. KINETIC STATE SPACE AND A COORDINATE FIREWALL

We work at a fixed spacetime event and resolve only the local massless momentum variables. The state space is

$$\mathcal{X} = (0, \infty) \times \mathbb{S}^2, \quad d\Pi = Cx^2 dx d\Omega, \quad C > 0, \quad (1)$$

where $x = E/(k_B T_{\text{ref}})$ is a direction-independent dimensionless energy. The factor x^2 belongs to the physical massless phase-space measure. The additional factor of x in the constraints below is an observable weight and is not absorbed into the entropy measure.

A. Statistics and convex entropy

We use the statistics label $\xi = 0, +1, -1$ for Maxwell–Boltzmann (MB), Bose–Einstein (BE), and Fermi–Dirac (FD) occupations, respectively, and write

$$\Phi_0(z) = e^{-z}, \quad \Phi_\xi(z) = \frac{1}{e^z - \xi} \quad (\xi = \pm 1). \quad (2)$$

The convex negative-entropy densities are

$$h_0(f) = f \ln f - f, \quad (3)$$

$$h_{+1}(f) = f \ln f - (1+f) \ln(1+f), \quad (4)$$

$$h_{-1}(f) = f \ln f + (1-f) \ln(1-f), \quad (5)$$

with physical domains $f > 0$ for MB and BE and $0 < f < 1$ for FD. They satisfy

$$h'_\xi(f) = \ln \frac{f}{1 + \xi f}, \quad h''_\xi(f) = \frac{1}{f(1 + \xi f)} > 0. \quad (6)$$

Thus

$$H_\xi[f] = \int_{\mathcal{X}} h_\xi(f) d\Pi, \quad S_\xi[f] = -k_B H_\xi[f], \quad (7)$$

is strictly convex in f on every convex admissible set, while S_ξ is strictly concave. For later use, the associated Bregman divergence is

$$D_{H_\xi}(f||g) = H_\xi[f] - H_\xi[g] - \int_{\mathcal{X}} h'_\xi(g)(f - g) d\Pi. \quad (8)$$

No convexity of Eq. (8) in its second argument is assumed anywhere in this paper.

Maximum-entropy reconstruction from moments and the associated realizability domains are well established [1, 3–7]. Quantum-statistical maximum-entropy radiation closures are likewise not new [10–12]. Our use of these ingredients is specialized to finite angular energy data and to the response provenance of observables evaluated on the selected representative.

Definition 1 (Primal admissible class). Let $\mathfrak{A}_\xi$ be the measurable occupations on $\mathcal{X}$ taking values in the closed physical domain ($f \geq 0$ for MB and BE, $0 \leq f \leq 1$ for FD), with $H_\xi[f]$ finite after continuous extension of h_ξ to the endpoints and with every retained moment finite. For a target m , let

$$\mathfrak{F}_{\xi, \Lambda}(m) = \{f \in \mathfrak{A}_\xi : M_i[f] = m_i, \ 1 \leq i \leq N_\Lambda\}. \quad (9)$$

The representative theorem concerns targets for which this class has an interior positive minimizer; it makes that condition explicit through the dual cone below.

B. Finite angular energy data

Let $\{Z_i\}_{i=1}^{N_\Lambda}$ be a real orthonormal spherical-harmonic basis of

$$V_\Lambda = \text{span}\{Z_{\ell a} : 0 \leq \ell \leq \Lambda\}, \quad N_\Lambda = (\Lambda + 1)^2, \quad (10)$$

including the positive constant mode. The retained energy moments are

$$M_i[f] = \int_{\mathcal{X}} x Z_i(\hat{e}) f(x, \hat{e}) d\Pi = C \int_{\mathbb{S}^2} Z_i \int_0^\infty x^3 f dx d\Omega. \quad (11)$$

For prescribed data $m = (m_i)$, the primal problem is

$$\min_{f \in \mathfrak{F}_{\xi, \Lambda}(m)} H_\xi[f]. \quad (12)$$

The constraints are linear in f , so strict convexity gives uniqueness once a feasible minimizer has been constructed. Existence for the interior data considered here will follow from the finite-dimensional dual problem rather than from a generic assertion about entropy closures.

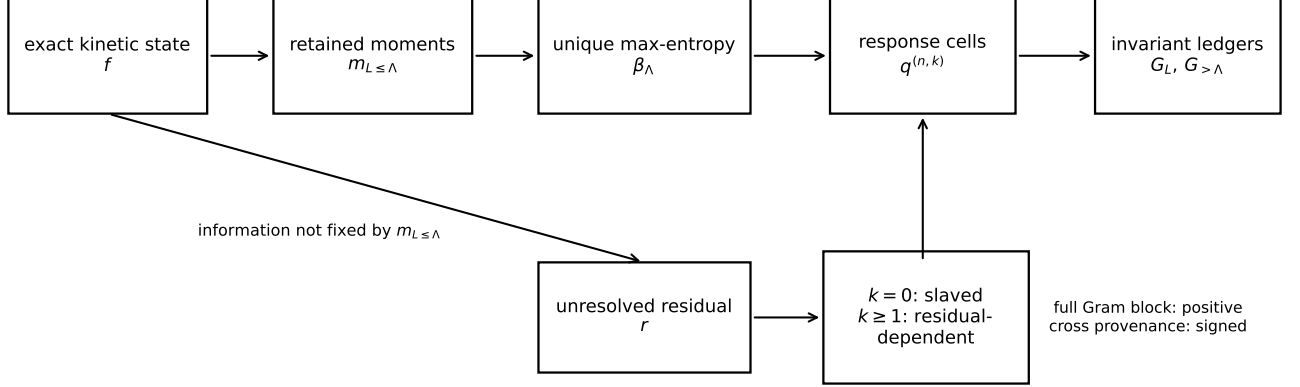


FIG. 1. Information flow in the construction. Retained angular energy moments select a unique maximum-entropy representative. Nonlinear observable terms generated from that representative have insertion count $k = 0$ and are algebraically slaved. Residual information not fixed by the moment vector enters through $k \geq 1$ residual-dependent terms. Shell contraction produces $SO(3)$ -invariant Gram ledgers.

C. Natural, response, and observable coordinates

The Euler–Lagrange natural parameter is an inverse-temperature field,

$$\beta_b(\hat{e}) = \sum_{i=1}^{N_\Lambda} b_i Z_i(\hat{e}), \quad f_{\xi,b}(x, \hat{e}) = \Phi_\xi[x\beta_b(\hat{e})]. \quad (13)$$

The finite maximum-entropy manifold is therefore defined by harmonic truncation of β , not of the temperature itself. To organize nonlinear response, choose a constant $\beta_0 > 0$ and define

$$v = \frac{\beta}{\beta_0} - 1, \quad \Theta = \frac{\beta_0}{\beta} = (1 + v)^{-1}, \quad u = \ln \Theta = -\ln(1 + v). \quad (14)$$

The field v has the same angular bandwidth as β and will be the response coordinate in Sec. VI; Θ is the physical temperature ratio, and u is the logarithmic temperature.

The full positive-field charts are equivalent, but finite truncation does not commute with the nonlinear transformations in Eq. (14). In general,

$$\Pi_{\leq \Lambda} \beta, \quad \beta_0 e^{-\Pi_{\leq \Lambda} u}, \quad \frac{\beta_0}{\Pi_{\leq \Lambda} \Theta} \quad (15)$$

are distinct whenever the latter two expressions are admissible. We therefore maintain a coordinate firewall throughout: entropy projection is performed in the natural β chart, response order is counted in v , and temperature semantics are expressed through Θ . The numerical consequences of replacing the natural chart by separately moment-matched finite u or Θ charts are reported in Sec. V.

III. STATISTICS-UNIFORM MAXIMUM-ENTROPY REPRESENTATIVES

A. Natural-parameter and dual moment cones

Define the closed nonnegative cone and its interior by

$$K_\Lambda = \{b \in \mathbb{R}^{N_\Lambda} : \beta_b(\hat{e}) \geq 0 \ \forall \hat{e} \in \mathbb{S}^2\}, \quad (16)$$

$$\mathcal{B}_\Lambda = \text{int } K_\Lambda = \{b : \beta_b(\hat{e}) > 0 \ \forall \hat{e} \in \mathbb{S}^2\}. \quad (17)$$

The cone K_Λ is pointed and full dimensional. Its dual moment cone is

$$\mathcal{R}_\Lambda = K_\Lambda^* = \{m : c \cdot m \geq 0 \text{ for all } c \in K_\Lambda\}. \quad (18)$$

Every moment vector of a state in Definition 1 belongs to $\mathcal{R}_\Lambda$, because its angular energy density is nonnegative. The theorem below shows conversely that every point of $\text{int } \mathcal{R}_\Lambda$ is realized by a unique positive one-field representative. We therefore identify $\text{int } \mathcal{R}_\Lambda$ with the interior realizable moment set for the present problem. No representative on $\partial \mathcal{R}_\Lambda$ is claimed.

B. Convex conjugates and radial factorization

For the natural dual variable $y < 0$, the convex conjugates of Eqs. (3)–(5) are

$$\begin{aligned} h_0^*(y) &= e^y, \\ h_{+1}^*(y) &= -\ln(1 - e^y), \\ h_{-1}^*(y) &= \ln(1 + e^y). \end{aligned} \quad (19)$$

They obey

$$(h_\xi^*)'(y) = \frac{1}{e^{-y} - \xi}, \quad (h_\xi^*)''(y) = f(1 + \xi f) > 0. \quad (20)$$

The dual potential associated with Eq. (12) is

$$A_{\xi,\Lambda}(b) = C \int_{\mathbb{S}^2} \int_0^\infty x^2 h_\xi^*[-x\beta_b(\hat{e})] dx d\Omega. \quad (21)$$

A radial rescaling $s = x\beta_b$ gives the exact factorization

$$A_{\xi,\Lambda}(b) = a_\xi C \int_{\mathbb{S}^2} \beta_b^{-3} d\Omega, \quad (22)$$

where

$$a_0 = 2, \quad a_{+1} = 2\zeta(4), \quad a_{-1} = 2(1 - 2^{-3})\zeta(4). \quad (23)$$

These Bose and Fermi integrals are standard [21]; their role here is to isolate all statistics dependence in a positive scalar.

The derivatives of Eq. (22) are

$$M_{\xi,i}(b) = -\partial_{b_i} A_{\xi,\Lambda}(b) = 3a_\xi C \int_{\mathbb{S}^2} Z_i \beta_b^{-4} d\Omega, \quad (24)$$

$$[\nabla^2 A_{\xi,\Lambda}(b)]_{ij} = 12a_\xi C \int_{\mathbb{S}^2} Z_i Z_j \beta_b^{-5} d\Omega. \quad (25)$$

For every nonzero $c \in \mathbb{R}^{N_\Lambda}$,

$$c^T \nabla^2 A_{\xi,\Lambda} c = 12a_\xi C \int_{\mathbb{S}^2} \left(\sum_i c_i Z_i \right)^2 \beta_b^{-5} d\Omega > 0. \quad (26)$$

Hence the dual potential is strictly convex on $\mathcal{B}_\Lambda$.

Lemma 2 (Planck-edge regularity). *For the zero-chemical-potential BE family, Eq. (21) and its first two parameter derivatives are locally integrable at $x = 0$. On every compact positive- β subcone they admit integrable dominators, and Eq. (22) defines a smooth potential on $\mathcal{B}_\Lambda$.*

The low-energy estimates and the boundary argument are given in Appendix A. The lemma concerns the absolutely continuous Planck edge. It does not address fixed-number Bose condensation.

Theorem 3 (Interior maximum-entropy representative). *For each $\xi \in \{0, +1, -1\}$, the moment map*

$$M_\xi = -\nabla A_{\xi,\Lambda} : \mathcal{B}_\Lambda \longrightarrow \text{int } \mathcal{R}_\Lambda \quad (27)$$

is a smooth diffeomorphism. Thus $\text{int } \mathcal{R}_\Lambda$ is precisely the interior realizable moment set for the primal class in Definition 1. For every $m \in \text{int } \mathcal{R}_\Lambda$, the inverse map determines

$$f_{\xi,m}^*(x, \hat{e}) = \Phi_\xi[x\beta_{M_\xi^{-1}(m)}(\hat{e})], \quad (28)$$

which is the unique minimizer of H_ξ among admissible distributions satisfying $M_i[f] = m_i$.

Proof sketch. Equation (26) gives local invertibility and dual uniqueness. The inverse-cubic factor in Eq. (22) diverges when a bounded positive harmonic field approaches zero, while $b \cdot m$ is coercive along unbounded sequences in K_Λ for $m \in \text{int } \mathcal{R}_\Lambda$. Thus

$$\Psi_{\xi,m}(b) = A_{\xi,\Lambda}(b) + b \cdot m \quad (29)$$

has a unique interior minimum, whose critical equation is $M_\xi(b) = m$. For any other feasible f , the linear term in $D_{H_\xi}(f \| f_{\xi,m}^*)$ vanishes because $h'_\xi(f_{\xi,m}^*) = -x\beta_b$ lies in the span of the retained constraints. Strict convexity then gives the unique primal minimum. Appendix A supplies the complete cone, barrier, and Planck-edge argument. $\square$

Corollary 4 (Statistics-uniform angular geometry). *After dividing the moment target by a_ξ , all three statistics have the same natural-parameter cone, normalized moment map, boundary barrier, and Hessian condition number. This does not identify arbitrary unnormalized physical data across statistics.*

Proposition 5 (Gram–Hessian identity). *For a parameter perturbation δb , let*

$$\delta f = -x f_{\xi,b}(1 + \xi f_{\xi,b}) \delta \beta, \quad \delta \beta = \sum_i \delta b_i Z_i. \quad (30)$$

Then the second Fréchet variation of the entropy satisfies

$$\delta^2 S_\xi[f_{\xi,b}](\delta f, \delta f) = -k_B \delta b^T \nabla^2 A_{\xi,\Lambda}(b) \delta b. \quad (31)$$

Thus the same positive matrix is the dual Hessian, the Fisher Gram matrix of the tangent vectors, and the negative entropy Hessian restricted to the manifold.

This triple role is inherited from exponential-family geometry; the specialized content of Theorem 3 is the global interior angular-cone result and its later use in observable provenance.

IV. CLASS-RELATIVE MINIMALITY AND ITS LIMITS

Theorem 3 selects a one-field inverse-temperature family from energy moments. We now ask a narrower question: among finite monomial natural-parameter families, which ones preserve the same finite positive linear high-energy slope on an open full-dimensional parameter set? Higher-order entropy closures commonly use polynomial exponents and can be mathematically valid [4–6, 8]; the result below does not deny their existence.

Let $\mathcal{W} \subset \mathbb{Z}_{\geq 0}$ be finite and nonempty. For every $m \in \mathcal{W}$, let $\lambda_m \in \bar{V}_{\Lambda_m} \subset C(\mathbb{S}^2)$ and define

$$g_\lambda(E, \hat{e}) = \sum_{m \in \mathcal{W}} \lambda_m(\hat{e}) E^m, \quad f_\lambda = \Phi_\xi(g_\lambda). \quad (32)$$

TABLE I. One-field statistics included in Theorem 3. The BE entry is the absolutely continuous Planck sector; a singular condensate measure and an independent number constraint are outside the theorem.

Statistics	occupation domain	$h_\xi^*(y)$ for $y < 0$	a_ξ	low-energy statement
MB ($\xi = 0$)	$f > 0$	e^y	2	regular
Planck BE ($\xi = +1$)	$f > 0$	$-\ln(1 - e^y)$	$2\zeta(4)$	integrable Planck edge
zero- μ FD ($\xi = -1$)	$0 < f < 1$	$\ln(1 + e^y)$	$2(1 - 2^{-3})\zeta(4)$	regular

The linear-Wien-tail class is specified by

$$0 < \lim_{E \rightarrow \infty} \frac{g_\lambda(E, \hat{e})}{E} < \infty \quad \text{for every } \hat{e} \in \mathbb{S}^2. \quad (33)$$

This high-energy condition is distinct from statistics-specific admissibility at low energy.

Theorem 6 (Robust class-relative minimality). *A nonempty open set in the full coefficient space*

$$\mathcal{P}_\mathcal{W} = \prod_{m \in \mathcal{W}} V_{\Lambda_m} \quad (34)$$

can consist of exponents satisfying Eq. (33) if and only if

$$1 \in \mathcal{W} \subseteq \{0, 1\}. \quad (35)$$

If any degree $m \geq 2$ is present, every class-preserving member obeys $\lambda_m \equiv 0$ for all such degrees and lies in a closed subspace of codimension

$$\sum_{\substack{m \in \mathcal{W} \\ m \geq 2}} \dim V_{\Lambda_m} > 0, \quad (36)$$

which has empty interior in $\mathcal{P}_\mathcal{W}$.

Proof sketch. For each fixed direction, divide the polynomial $g_\lambda(E, \hat{e})$ by E . A finite positive limit requires every coefficient of degree $m \geq 2$ to vanish and the coefficient of E to be positive. Continuity of the angular multiplier fields promotes almost-everywhere vanishing to pointwise vanishing. The resulting high-degree kernel is a closed positive-codimension linear subspace whenever a higher weight is present. Conversely, the positive cone $\lambda_1 > 0$ is open for $\mathcal{W} = \{1\}$ and remains open after adding an intercept for $\mathcal{W} = \{0, 1\}$, subject to separate low-energy admissibility. Appendix B gives the full proof. $\square$

The condition includes the linear weight. The set $\mathcal{W} = \{0\}$ has $g(E)/E \rightarrow 0$ and therefore does not satisfy Eq. (33). For $\mathcal{W} = \{0, 1\}$ one may write

$$g(E, \hat{e}) = \beta(\hat{e})E - \alpha(\hat{e}). \quad (37)$$

The high-energy requirement is $\beta > 0$. MB and FD allow arbitrary finite α , while BE requires $\alpha \leq 0$ for pointwise admissibility, with $\alpha = 0$ the integrable Planck edge. Equation (37) is only a family-level class-preservation statement here; no global two-field moment-map diffeomorphism is claimed.

Theorem 6 is not a solution-nonexistence theorem. Two countercontrols make the distinction explicit. First, the enlarged MB family

$$f_{\beta, \gamma}(E) = e^{-\beta E - \gamma E^2} \quad (38)$$

contains an ordinary Teff state at $\gamma = 0$. Second, for $\beta, \gamma > 0$, Eq. (38) is the unique maximum-entropy state for its own energy and quadratic-moment target, yet

$$\frac{-\ln f_{\beta, \gamma}(E)}{E} = \beta + \gamma E \rightarrow \infty. \quad (39)$$

It exists but leaves the linear-Wien-tail class. The correct conclusion is class exit, not failure of maximum entropy or intrinsic physical invalidity.

V. EXACT ENTROPY INEQUALITIES ACROSS ANGULAR RESOLUTION

The entropy and Pythagorean identities in this section are standard information-projection structures, recorded here in the specialized form made exact by angular energy-moment matching [2, 18]. Their role is to select and compare representatives that feed the response ledger below; the identities themselves are not claimed as the paper's methodological novelty. The representative theorem makes the specialization direct because the natural parameter of the lower-resolution state lies in the span of the retained moment constraints. No generic convexity of a Bregman divergence in its second argument, and no additional projection-regularity hypothesis, is required.

For an admissible exact state f , let f_Λ^* denote the unique representative whose moments agree with f for all real harmonics with $L \leq \Lambda$, and write

$$S_{\xi, \Lambda} = S_\xi[f_\Lambda^*]. \quad (40)$$

All statements in this section concern moment vectors in the interior domains of Theorem 3.

Theorem 7 (Exact moment-matched entropy gap). *If f and f_Λ^* share the retained energy moments, then*

$$S_\xi[f_\Lambda^*] - S_\xi[f] = k_B D_{H_\xi}(f \| f_\Lambda^*) \geq 0. \quad (41)$$

Equality holds if and only if $f = f_\Lambda^$ almost everywhere.*

Proof. The Bregman linear term is

$$\begin{aligned} & \int_{\mathcal{X}} h'_\xi(f_\Lambda^*)(f - f_\Lambda^*) d\Pi \\ &= - \int_{\mathcal{X}} x\beta_\Lambda(\hat{e})(f - f_\Lambda^*) d\Pi \\ &= - \sum_i b_i [M_i[f] - M_i[f_\Lambda^*]] = 0, \end{aligned} \quad (42)$$

where $\beta_\Lambda = \sum_i b_i Z_i$. Hence $D_{H_\xi}(f \| f_\Lambda^*) = H_\xi[f] - H_\xi[f_\Lambda^*]$, which is Eq. (41). Strict convexity of H_ξ gives the equality condition. $\square$

Corollary 8 (Nested entropy filtration). *For nested angular moment sets,*

$$S_{\xi,0} \geq S_{\xi,1} \geq \dots \geq S_{\xi,\Lambda} \geq S_{\xi,\Lambda+1} \geq S_\xi[f]. \quad (43)$$

The equality $S_{\xi,\Lambda} = S_{\xi,\Lambda+1}$ holds precisely when the coarser representative already reproduces the newly imposed shell and therefore coincides with $f_{\Lambda+1}^$.*

Proof. The feasible set at resolution $\Lambda + 1$ is contained in the feasible set at resolution Λ . Maximizing entropy over the smaller set cannot increase the maximum. The lower bound follows because the exact state is feasible at every finite resolution. Uniqueness gives the equality statement. $\square$

Proposition 9 (Adjacent-representative and moment-Pythagorean identities). *For adjacent interior representatives,*

$$S_{\xi,\Lambda} - S_{\xi,\Lambda+1} = k_B D_{H_\xi}(f_{\Lambda+1}^* \| f_\Lambda^*). \quad (44)$$

Consequently, every exact state sharing the full target hierarchy satisfies

$$D_{H_\xi}(f \| f_\Lambda^*) = D_{H_\xi}(f \| f_{\Lambda+1}^*) + D_{H_\xi}(f_{\Lambda+1}^* \| f_\Lambda^*). \quad (45)$$

Proof. The natural parameter $h'_\xi(f_\Lambda^*) = -x\beta_\Lambda$ contains only harmonics through Λ . The two adjacent representatives share all of those moments, so the linear term in $D_{H_\xi}(f_{\Lambda+1}^* \| f_\Lambda^*)$ vanishes exactly as in Eq. (42). Equation (44) follows after converting H_ξ to S_ξ . Combining it with Theorem 7 at the two resolutions gives Eq. (45). $\square$

Figure 2(a) shows an independent MB hierarchy. The exact entropy gap and a separately integrated Bregman divergence agree with maximum relative discrepancy 6.24×10^{-13} . A second radial rule reproduces the adjacent identity with maximum relative discrepancy 4.64×10^{-10} ; the theorem itself is statistics-uniform.

A. Finite-chart noncommutativity

The full positive fields β , $u = -\ln(\beta/\beta_0)$, and $\Theta = \beta_0/\beta$ are equivalent coordinates, but their finite harmonic

submanifolds are not. At fixed Λ , define

$$\begin{aligned} \mathcal{M}_\Lambda^{(\beta)} &= \{\Phi_\xi(x\beta_\Lambda) : \beta_\Lambda \in V_\Lambda, \beta_\Lambda > 0\}, \\ \mathcal{M}_\Lambda^{(u)} &= \{\Phi_\xi(x\beta_0 e^{-u_\Lambda}) : u_\Lambda \in V_\Lambda\}, \\ \mathcal{M}_\Lambda^{(\Theta)} &= \{\Phi_\xi(x\beta_0/\Theta_\Lambda) : \Theta_\Lambda \in V_\Lambda, \Theta_\Lambda > 0\}. \end{aligned} \quad (46)$$

These sets generally intersect but are not identical.

Proposition 10 (Natural-chart entropy dominance). *Let $f_\Lambda^* \in \mathcal{M}_\Lambda^{(\beta)}$ be the unique maximum-entropy representative for an interior moment vector. If a state $\tilde{f}$ in either of the other finite chart families matches the same complete retained moment vector, then*

$$S_\xi[f_\Lambda^*] - S_\xi[\tilde{f}] = k_B D_{H_\xi}(\tilde{f} \| f_\Lambda^*) \geq 0, \quad (47)$$

with strict inequality whenever $\tilde{f} \neq f_\Lambda^$.*

This proposition is analytic; numerics are needed only to establish that the separately moment-matched nonlinear chart states are in fact distinct in a representative configuration. For the $\Lambda = 2$ MB test in Fig. 2(b), all three chart states match the same moment vector with validation residual at most 1.04×10^{-13} , yet

$$\begin{aligned} S_{\beta-\text{ME}} - S_u &= 1.87 \times 10^{-4}, \\ S_{\beta-\text{ME}} - S_\Theta &= 7.45 \times 10^{-4}, \\ \|\beta_u - \beta_{\beta-\text{ME}}\|_2 &= 3.83 \times 10^{-3}, \\ \|\beta_\Theta - \beta_{\beta-\text{ME}}\|_2 &= 7.71 \times 10^{-3}. \end{aligned} \quad (48)$$

Direct coordinate truncations are a different comparison: they need not satisfy the complete retained constraint vector and therefore are not subject to Eq. (47).

VI. ANGULAR RESPONSE LEDGER

Let f_Λ^* be the maximum-entropy representative selected by the retained moments and write its inverse temperature as

$$\beta_\Lambda(\hat{e}) = \beta_0[1 + v_R(\hat{e})], \quad v_R \in V_\Lambda, \quad 1 + v_R > 0. \quad (49)$$

For a massless radial observable with homogeneity index $p > 0$, radial integration separates from the angular dependence and gives, up to a statistics-dependent positive constant,

$$J_p[v_R] = (1 + v_R)^{-p}. \quad (50)$$

For integer p and $\|v_R\|_\infty < 1$, the binomial series converges uniformly,

$$J_p[v_R] = \sum_{n=0}^{\infty} c_n^{(p)} v_R^n, \quad c_n^{(p)} = (-1)^n \frac{(p)_n}{n!}. \quad (51)$$

The finite-order identities below are polynomial statements and remain meaningful without summing the infinite series. Appendix C gives a uniform remainder bound for the resummed observable.

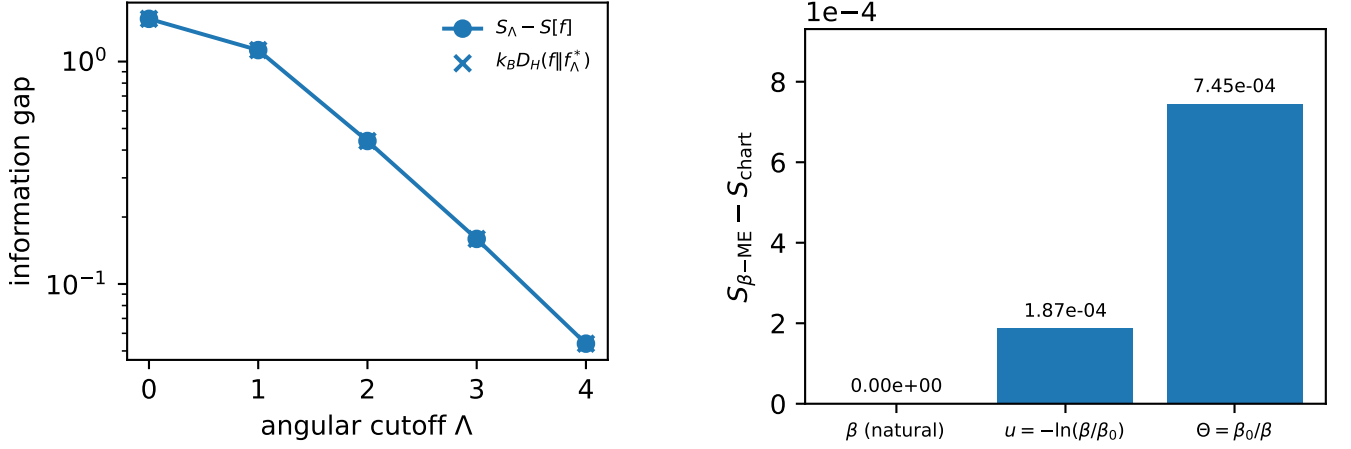


FIG. 2. Entropy projection and finite-chart comparison. (a) Exact entropy filtration for a positive nonaxisymmetric MB state: circles show $S_\Lambda - S[f]$ and crosses show the separately integrated $k_B D_H(f \| f_\Lambda^*)$. (b) Finite-chart noncommutativity after separate moment matching at $\Lambda = 2$: every retained moment is matched to 1.04×10^{-13} or better, but the nonlinear finite-chart states have lower entropy than the natural inverse-temperature representative.

A. Order-shell cells and hard support

Definition 11 (Slaved order-shell cell). The representative-generated order- n field and its output shell are

$$q^{(n,0)} = c_n^{(p)} v_R^n, \quad s_L^{(n,0)} = \Pi_L q^{(n,0)}. \quad (52)$$

The pair (n, L) records response order and output angular rank; the second superscript anticipates residual insertions below.

Theorem 12 (Hard support and parity). *If $v_R \in V_\Lambda$, then*

$$L > n\Lambda \implies s_L^{(n,0)} = 0. \quad (53)$$

If, in addition, v_R has definite parity $v_R(-\hat{e}) = (-1)^\sigma v_R(\hat{e})$ with $\sigma \in \{0, 1\}$, then

$$s_L^{(n,0)} = 0 \quad \text{unless} \quad L \equiv n\sigma \pmod{2}. \quad (54)$$

The ceiling in Eq. (53) is a necessary support condition, not a guarantee that every allowed shell is populated.

Proof. The product of harmonics of ranks ℓ_1 and ℓ_2 contains only ranks satisfying the Clebsch-Gordan triangle bound $L \leq \ell_1 + \ell_2$ and the parity rule $\ell_1 + \ell_2 + L$ even. Iteration through n factors gives $L \leq \sum_i \ell_i \leq n\Lambda$. Definite parity of the product gives Eq. (54). Appendix C records the Gaunt-coefficient proof and the real-basis translation. $\square$

Figure 3 shows the actual populated cells for a nonaxisymmetric odd-parity field in V_3 . The triangular ceiling and the alternating parity zeros are both visible; allowed cells can nevertheless be absent because of coefficient cancellations or missing input ranks.

B. Representative-relative residual insertions

The exact kinetic state need not lie on the representative manifold, and an observable can depend on residual data that are not functions of v_R . Let r denote one such residual coordinate; it may be an omitted angular field, an off-manifold spectral coefficient, or an observable-specific residual such as w_p in Eq. (85). Introduce the bivariate generating path

$$\hat{\mathcal{J}}(t, s) = \mathcal{J}[tv_R, sr]. \quad (55)$$

The zero-amplitude reference is $(t, s) = (0, 0)$, the selected representative is $(1, 0)$, and the residual-augmented description is $(1, 1)$. We assume analyticity in a polydisk containing these points when a resummed expansion is

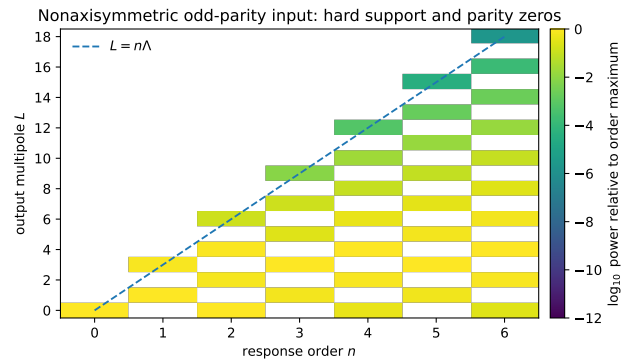


FIG. 3. Order-shell support for an odd-parity nonaxisymmetric representative field $v_R \in V_3$ and the $p = 4$ observable. Color shows shell power relative to the largest shell at each order. Blank cells are parity-forbidden or below 10^{-14} of the order maximum. No power occurs beyond the hard ceiling $L = 3n$.

used. Fixed-order polynomial identities require only the corresponding derivatives.

Definition 13 (Insertion-resolved amplitude ledger). For the generating path in Eq. (55), define

$$q^{(n,k)}[v_R, r] = \frac{1}{n!k!} \left. \frac{\partial^{n+k} \hat{\mathcal{J}}}{\partial t^n \partial s^k} \right|_{t=s=0}, \quad (56)$$

$$s_L^{(n,k)} = \Pi_L q^{(n,k)}. \quad (57)$$

The index n is representative-amplitude composition order, not a tangent expansion about f_Λ^* . Terms with $k = 0$ are *slaved* to the selected representative. Terms with $k \geq 1$ are called *residual-dependent*: their value requires information not fixed by that moment vector. The label is about data provenance and does not assert dynamical independence.

This terminology is about provenance, not output rank. A slaved term can populate $L > \Lambda$ through nonlinear composition, while a residual-dependent term may contribute to a low shell. The residual need not be a pure high- L coordinate; the equal-information construction in Sec. VIII is off the Teff manifold in its radial shape. An arbitrary residual need not have finite angular bandwidth, so Theorem 12 is not asserted for the $k \geq 1$ sector without additional hypotheses.

Remark 14 (Coordinate status of the insertion ledger). The labels (n, k) are defined relative to the fixed response coordinate v_R , residual coordinate r , and scaling path (t, s) in Definition 13. A nonlinear reparameterization can redistribute terms among cells even though the exact observable and its summed shell coefficients are unchanged. The invariance proved below is under common $SO(3)$ rotations and orthonormal basis changes within an irreducible shell, not under arbitrary changes of response coordinates or response-label basis.

Remark 15 (Finite multi-residual extension). For finitely many residual coordinates $\mathbf{r} = (r_1, \dots, r_R)$, replace Eq. (55) by $\hat{\mathcal{J}}(t, \mathbf{s}) = \mathcal{J}[tv_R, s_1 r_1, \dots, s_R r_R]$ and define

$$q^{(n, \mathbf{k})} = \frac{1}{n! \prod_{j=1}^R k_j!} \left. \frac{\partial^{n+|\mathbf{k}|} \hat{\mathcal{J}}}{\partial t^n \prod_{j=1}^R \partial s_j^{k_j}} \right|_{t=s=0}. \quad (58)$$

The slaved sector has $|\mathbf{k}| = 0$ and the residual-dependent sector has $|\mathbf{k}| \geq 1$. This is an exact finite- R extension. An infinite-dimensional residual space additionally requires a chosen basis and convergence control; no canonical residual basis or complete residual-space geometry is claimed here.

For the additive scalar realization

$$\mathcal{J}_p[v_R, r] = (1 + v_R + r)^{-p}, \quad (59)$$

Definition 13 gives

$$q^{(n, \mathbf{k})} = (-1)^{n+k} \frac{(p)_{n+k}}{n!k!} v_R^n r^{\mathbf{k}}. \quad (60)$$

Equivalently, at fixed total degree $d = n + k$, this is the binomial split of $c_d^{(p)}(v_R + r)^d$. The multiplicative residual family in Eq. (86) instead has $q^{(n,0)} = a_n^{(p)} v_R^n$ and $q^{(n,1)} = a_n^{(p)} v_R^n w_p$; the same insertion definition applies without forcing every residual into the inverse-temperature coordinate.

Proposition 16 (Cutoff-relative provenance). *Let the same exact observable admit decompositions $\mathcal{J} = \sum_{n,k} q_\Lambda^{(n,k)}$ and $\mathcal{J} = \sum_{n,k} q_{\Lambda'}^{(n,k)}$ relative to two nested retained descriptions with $\Lambda' < \Lambda$ or $\Lambda' > \Lambda$. The exact observable and its total shell coefficients are independent of this bookkeeping choice, but the division into $k = 0$ and $k \geq 1$ sectors need not be. In particular, information absorbed into the larger representative is reclassified from residual-dependent to slaved.*

The proposition is immediate from Definition 13: the residual coordinate is defined relative to the chosen representative. The numerical cutoff-translation control in Sec. IX uses an exact V_2 Teff state. At $\Lambda = 1$ it has nonzero residual-dependent power; at $\Lambda = 2$ the state itself is the representative, the residual vanishes, and the same finite-order observable is entirely slaved.

An ordinary harmonic expansion retains only the sum $\Pi_L \mathcal{J}$. The response ledger retains, before that sum, which response order and which residual insertion generated the shell. That additional provenance is the information used by the Gram construction below.

VII. SHELL-CONTRACTED GRAM LEDGERS AND THE LOCAL FISHER BRIDGE

Positive semidefiniteness of a Gram matrix and the Parseval staircase are standard Hilbert-space facts. We record them because contracting only after retaining the insertion labels produces the rotation-invariant provenance object used by the paper-specific ledger; no novelty is claimed for Gram positivity or Parseval's identity in isolation.

Fix a finite set $\mathcal{A}$ of response labels $a = (n, k)$, or an infinite set for which the expansion converges in $L^2(\mathbb{S}^2)$. Write $s_L^a = \Pi_L q^a$.

A. Shell and discarded-block Gram matrices

Definition 17 (Shell Gram ledger). For each irreducible shell L , define

$$G_L^{ab} = \langle s_L^a, s_L^b \rangle_{L^2(\mathbb{S}^2)}, \quad a, b \in \mathcal{A}. \quad (61)$$

For an angular cutoff Λ , define the discarded block

$$G_{>\Lambda} = \sum_{L > \Lambda} G_L. \quad (62)$$

Theorem 18 ($SO(3)$ -invariant positive Gram structure). *Every G_L and $G_{>\Lambda}$ is Hermitian and positive semidefinite. They are independent of the orthonormal basis chosen within a shell and invariant under a common rotation of all generated fields. In a real harmonic basis they are real symmetric.*

Proof. For any coefficient vector $z = (z_a)$,

$$z^\dagger G_L z = \left\| \sum_a z_a s_L^a \right\|_2^2 \geq 0. \quad (63)$$

A basis change within the L shell and the action of a rotation are unitary on that irreducible subspace, so they preserve every inner product in Eq. (61). Summing positive-semidefinite matrices proves the discarded-block statement. $\square$

Mode coefficients transform covariantly and therefore depend on orientation; G_L is the first level in this contraction chain that has discarded that orientation while retaining order and insertion interference. It is not a complete invariant of the generated field: phase and orientation information within each irreducible shell have deliberately been removed. The response labels $a = (n, k)$ are external bookkeeping labels rather than $SO(3)$ representation indices; a linear change of that label basis transforms G_L by congruence, and a nonlinear coordinate change can alter the cell decomposition as stated in Remark 14. The information-loss chain is

$$\{q^a\} \longrightarrow \{s_L^a\}_{L,a} \longrightarrow \{G_L\}_L \longrightarrow G_{>\Lambda}. \quad (64)$$

B. Exact discarded-content staircase

Let $\mathbf{1}$ denote the vector with one entry for every response label and suppose $\mathcal{J} = \sum_a q^a$ in L^2 . Define

$$\mathcal{L}_\Lambda[\mathcal{J}] = \mathbf{1}^\dagger G_{>\Lambda} \mathbf{1}. \quad (65)$$

Theorem 19 (Exact L^2 staircase). *The scalar ledger is exactly the discarded directional content,*

$$\mathcal{L}_\Lambda[\mathcal{J}] = \|Q_\Lambda \mathcal{J}\|_{L^2(\mathbb{S}^2)}^2, \quad (66)$$

and obeys

$$\mathcal{L}_\Lambda[\mathcal{J}] - \mathcal{L}_{\Lambda+1}[\mathcal{J}] = \|\Pi_{\Lambda+1} \mathcal{J}\|_2^2 \geq 0. \quad (67)$$

Proof. Equation (66) follows by expanding the norm of $Q_\Lambda \sum_a q^a$. The orthogonal decomposition $Q_\Lambda = \Pi_{\Lambda+1} + Q_{\Lambda+1}$ gives Eq. (67) by Parseval's identity. $\square$

The staircase is an exact L^2 statement about an angular observable. It is not, by itself, a kinetic-entropy identity.

C. Slaved, residual-dependent, and cross sectors

Define the aggregate shell fields

$$s_{L,\text{slv}} = \sum_{a:k(a)=0} s_L^a, \quad s_{L,\text{res}} = \sum_{a:k(a)\geq 1} s_L^a. \quad (68)$$

Then the full shell power decomposes exactly as

$$\|\Pi_L \mathcal{J}\|_2^2 = P_{L,\text{slv}} + P_{L,\text{res}} + P_{L,\times}, \quad (69)$$

$$P_{L,\text{slv}} = \|s_{L,\text{slv}}\|_2^2,$$

$$P_{L,\text{res}} = \|s_{L,\text{res}}\|_2^2,$$

$$P_{L,\times} = 2 \operatorname{Re} \langle s_{L,\text{slv}}, s_{L,\text{res}} \rangle. \quad (70)$$

The slaved and residual-dependent diagonal blocks are positive semidefinite. The cross block, and hence $P_{L,\times}$, can have either sign; positivity belongs to the complete Gram contraction, not to every provenance component separately.

Figure 4 verifies the provenance and invariance statements on a nonaxisymmetric representative-relative split. Shells $L = 0, 1$ are almost entirely representative-generated, whereas the residual dominates the higher shells. The normalized components sum to unity shell by shell, and thirty common random rotations change any active-shell Gram matrix only at floating-point level. The final panel displays the cutoff-relative reclassification of one fixed exact state: increasing the retained cutoff absorbs its residual into the representative without changing the total observable.

D. Local Fisher bridge and its boundary

Let g be an admissible kinetic distribution and let δf be such that $g + t\delta f$ remains admissible for $0 \leq t \leq 1$. Assume the third-order Taylor remainder is integrable. Then

$$D_{H_\xi}(g + \delta f \| g) = \frac{1}{2} \int_{\mathcal{X}} \frac{(\delta f)^2}{g(1 + \xi g)} d\Pi + O(\|\delta f\|^3). \quad (71)$$

This is the local Fisher bridge. It concerns a distribution residual measured with the entropy Hessian; Eq. (67) concerns an angular observable measured in unweighted L^2 .

At an isotropic Teff state $f_0(x) = \Phi_\xi(\beta_0 x)$, set $\beta = \beta_0(1 + \epsilon a)$ with $a \in L^\infty(\mathbb{S}^2)$. To first order,

$$\delta f = -\epsilon \beta_0 x f_0(1 + \xi f_0) a + O(\epsilon^2). \quad (72)$$

Rotational invariance then factorizes the quadratic form:

$$D_{H_\xi}(f_{\epsilon a} \| f_0) = \frac{\epsilon^2}{2} I_{\xi, \beta_0} \sum_{L \geq 0} \|\Pi_L a\|_2^2 + O(\epsilon^3), \quad (73)$$

$$I_{\xi, \beta_0} = C \beta_0^{-3} \int_0^\infty y^4 \Phi_\xi(y) [1 + \xi \Phi_\xi(y)] dy. \quad (74)$$

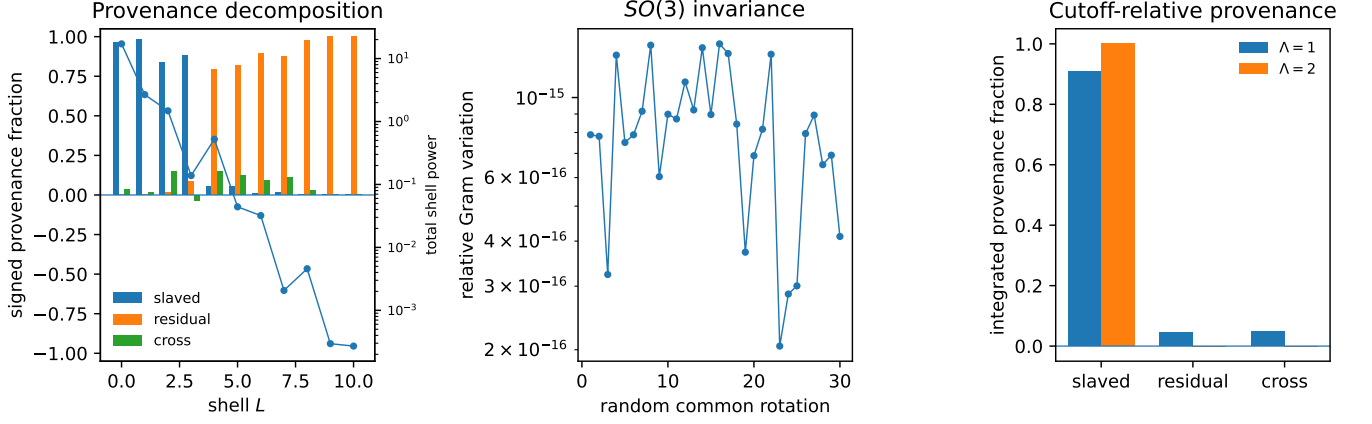


FIG. 4. Invariant provenance ledger for the additive $p = 4$ response through total order five. (a) Slaved, residual-dependent, and signed cross contributions, normalized by total power in each active shell. Their sum is exactly one; a cross contribution need not be positive. (b) Maximum relative change of the term-resolved shell Gram matrices under thirty common random $SO(3)$ rotations. (c) Integrated provenance fractions for the same exact V_2 state at cutoffs $\Lambda = 1$ and 2. At $\Lambda = 2$ the residual vanishes and all finite-order content is slaved, while the total shell-power vector agrees with the $\Lambda = 1$ classification to 3.37×10^{-16} relative.

The radial factors are

$$\begin{aligned} I_{0,\beta_0} &= 24C\beta_0^{-3}, \\ I_{+1,\beta_0} &= 24C\beta_0^{-3}\zeta(4), \\ I_{-1,\beta_0} &= 24C\beta_0^{-3}(1 - 2^{-3})\zeta(4). \end{aligned} \quad (75)$$

Away from isotropy the Fisher weight depends on direction, so ordinary harmonic shells need not diagonalize the quadratic form. One must then retain the full weighted Gram matrix. Likewise, the unweighted observable Gram ledger becomes an entropy proxy only after the residual entering that ledger has been explicitly related to the distribution-space Fisher residual; no global equality is claimed.

VIII. EQUAL RETAINED INFORMATION AND DISTINCT RESIDUAL-DEPENDENT CONTENT

The entropy filtration says exactly how the representative changes when more moments are supplied. It does not imply that a fixed retained moment vector determines every other observable of the exact state. We now give an analytic positive counterfamily that separates these two statements. The construction is MB-specific, which is sufficient for the information-theoretic negative result.

Let $\beta \in V_{\Lambda_0}$ be strictly positive, set $y = \beta(\hat{e})x$, and choose $g \in L^\infty(\mathbb{S}^2)$ with $\|g\|_\infty \leq 1$. Define

$$f_{\varepsilon,g}(x, \hat{e}) = e^{-y} [1 + \varepsilon e^{-y}(y - 2)g(\hat{e})], \quad 0 \leq \varepsilon < \frac{1}{2}. \quad (76)$$

Lemma 20 (Global positivity). *Every state in Eq. (76) is strictly positive on $(0, \infty) \times \mathbb{S}^2$.*

Proof. For $q(y) = e^{-y}(y - 2)$ with $y \geq 0$, one has

$$-2 \leq q(y) \leq e^{-3}. \quad (77)$$

The minimum occurs at $y = 0$ and the maximum at $y = 3$. Hence $1 + \varepsilon q(y)g(\hat{e}) \geq 1 - 2\varepsilon > 0$. $\square$

Suppressing common phase-space constants, define the radial observable family

$$\mathcal{O}_p[f](\hat{e}) = \int_0^\infty x^{p-1} f(x, \hat{e}) dx, \quad p > 0. \quad (78)$$

Here $p = 3$ has the radial weight of a number-density sector and $p = 5$ is a *higher-radial-weight observable* carrying one additional energy power. It is not a reaction rate: no opacity, matrix element, threshold, or microphysical collision kernel is included in $\mathcal{O}_5$.

Lemma 21 (Exact radial fingerprint). *For Eq. (76),*

$$\mathcal{O}_p[f_{\varepsilon,g}] = \Gamma(p)\beta^{-p} [1 + \varepsilon c_p g], \quad c_p = \frac{p-4}{2^{p+1}}. \quad (79)$$

In particular,

$$c_3 = -\frac{1}{16}, \quad c_4 = 0, \quad c_5 = \frac{1}{64}. \quad (80)$$

Proof. The base integral is $\Gamma(p)\beta^{-p}$. In the residual term, $y = \beta x$ gives

$$\begin{aligned} & \int_0^\infty x^{p-1} e^{-2\beta x} (\beta x - 2) dx \\ &= \beta^{-p} \left[\frac{\Gamma(p+1)}{2^{p+1}} - \frac{2\Gamma(p)}{2^p} \right] = \Gamma(p)\beta^{-p} \frac{p-4}{2^{p+1}}. \end{aligned} \quad (81)$$

$\square$

The cancellation at $p = 4$ is pointwise in direction, not merely after angular projection. Since the retained energy density in Eq. (11) is proportional to $\mathcal{O}_4$, it removes the residual from every angular energy moment.

Theorem 22 (Equal information, common representative, and residual nonidentifiability). *Let g_A and g_B satisfy the hypotheses above. Then*

$$\mathcal{O}_4[f_{\varepsilon, g_A}](\hat{e}) = \mathcal{O}_4[f_{\varepsilon, g_B}](\hat{e}) = 6\beta(\hat{e})^{-4} \quad (82)$$

for every direction. Consequently, for every $\Lambda \geq \Lambda_0$, the two exact states have the same complete retained energy vector and the same unique maximum-entropy representative,

$$f_{\Lambda}^{\star(A)} = f_{\Lambda}^{\star(B)} = e^{-\beta x}. \quad (83)$$

All observable terms generated solely from this representative are therefore identical. Nevertheless, for $p \neq 4$,

$$\mathcal{O}_p[f_{\varepsilon, g_A}] - \mathcal{O}_p[f_{\varepsilon, g_B}] = \varepsilon c_p \Gamma(p) \beta^{-p} (g_A - g_B), \quad (84)$$

so the retained energy data do not determine the residual-dependent number-like ($p = 3$) or higher-radial-weight ($p = 5$) observables. No rate interpretation is attached to the latter: it contains no opacity, matrix element, threshold, or microphysical collision kernel.

Proof. Equation (82) follows from $c_4 = 0$. The state $e^{-\beta x}$ is a feasible interior stationary state for the shared moment vector at every cutoff containing β ; Theorem 3 makes it the unique representative. Equation (84) is the difference of Eq. (79) for the two residual fields. $\square$

To connect the theorem to the response ledger, write

$$\beta = \beta_0(1 + v), \quad w_p = \varepsilon c_p g. \quad (85)$$

After division by $\Gamma(p)\beta_0^{-p}$,

$$\begin{aligned} \hat{\mathcal{O}}_p &= (1 + v)^{-p} (1 + w_p) \\ &= \sum_{n \geq 0} a_n^{(p)} v^n + \sum_{n \geq 0} a_n^{(p)} v^n w_p, \quad a_n^{(p)} = (-1)^n \frac{(p)_n}{n!}. \end{aligned} \quad (86)$$

The first sum is the representative-generated, or slaved, sector. The second contains one explicit residual insertion and is residual-dependent relative to the representative. For $p = 4$, $w_4 = 0$ and the entire residual-dependent sector vanishes. For $v \in V_{\Lambda_0}$, each slaved order obeys the hard support $L \leq n\Lambda_0$, while the residual insertion may carry independent angular support.

For the numerical realization in Fig. 5, $\Lambda_0 = 2$ and the common inverse-temperature field is non-axisymmetric. The bounded residuals are chosen from a normalized $L = 4$ real harmonic for state A and a normalized $L = 5$ real harmonic for state B, using the addition-theorem bound to guarantee $|g| \leq 1$. Independent solves return representatives that differ by only 1.08×10^{-14} , while

every slaved Gram block agrees to machine precision. At residual order $n = 0$, state A carries residual-dependent $L = 4$ power and state B residual-dependent $L = 5$ power; the wrong-shell leakage is below 2×10^{-34} . The $p = 3$ and $p = 5$ insertions have opposite signs before Gram contraction, and the $p = 4$ residual-dependent coefficients vanish exactly.

The information cost also depends on how residual amplitudes are normalized. In the normalization used for Fig. 5, the addition-theorem L^∞ bounds give unequal quadratic norms. Indeed, the originally quoted ratio $D_H^A/D_H^B = 1.222027$ tracks the Fisher-norm ratio 1.222025; it is therefore not an intrinsic comparison of residual shape. Expanding the MB Bregman cost in the normalization used here gives

$$\begin{aligned} D_H(f_{\varepsilon, g} \| f_\beta) &= \frac{4}{81} \varepsilon^2 \int_{\mathbb{S}^2} \beta^{-3} g^2 d\Omega \\ &+ \frac{41}{3072} \varepsilon^3 \int_{\mathbb{S}^2} \beta^{-3} g^3 d\Omega + O(\varepsilon^4). \end{aligned} \quad (87)$$

After matching the leading Fisher norm $\int \beta^{-3} g^2 d\Omega = 1$, the same $\varepsilon = 0.25$ comparison becomes

$$D_H^A = 3.09035 \times 10^{-3}, \quad D_H^B = 3.09141 \times 10^{-3}, \quad (88)$$

with ratio 0.999655. The remaining absolute difference scales as $\varepsilon^{3.03}$ over $0.025 \leq \varepsilon \leq 0.16$, consistent with the cubic angular-structure term in Eq. (87). The equal-information theorem is unaffected; the correction is that information cost is amplitude- and metric-normalization dependent.

A. Physical anchor: an energy-matched two-temperature mixture

A familiar off-manifold state is a positive mixture of equilibrium spectra at different temperatures [14, 15]. Let $T_\star = \beta^{-1}$ and, for a bounded angular mixing field $|s(\hat{e})| < 1$, write $T_\pm = T_\pm(\hat{e})$ and define

$$f_{\text{mix}}(x, \hat{e}) = \frac{1}{2} e^{-x/T_+} + \frac{1}{2} e^{-x/T_-}. \quad (89)$$

$$\begin{aligned} T_\pm &= T_\star a(s)(1 \pm s), \\ a(s) &= (1 + 6s^2 + s^4)^{-1/4}. \end{aligned} \quad (90)$$

Its radial moments are

$$\begin{aligned} \mathcal{O}_p[f_{\text{mix}}] &= \Gamma(p) T_\star^p R_p(s), \\ R_p(s) &= a(s)^p \frac{(1+s)^p + (1-s)^p}{2}. \end{aligned} \quad (91)$$

By construction $R_4(s) = 1$ pointwise, so every angular energy moment and the maximum-entropy representative are unchanged. Nevertheless,

$$R_3(s) = 1 - \frac{3}{2} s^2 + O(s^4), \quad R_5(s) = 1 + \frac{5}{2} s^2 + O(s^4), \quad (92)$$

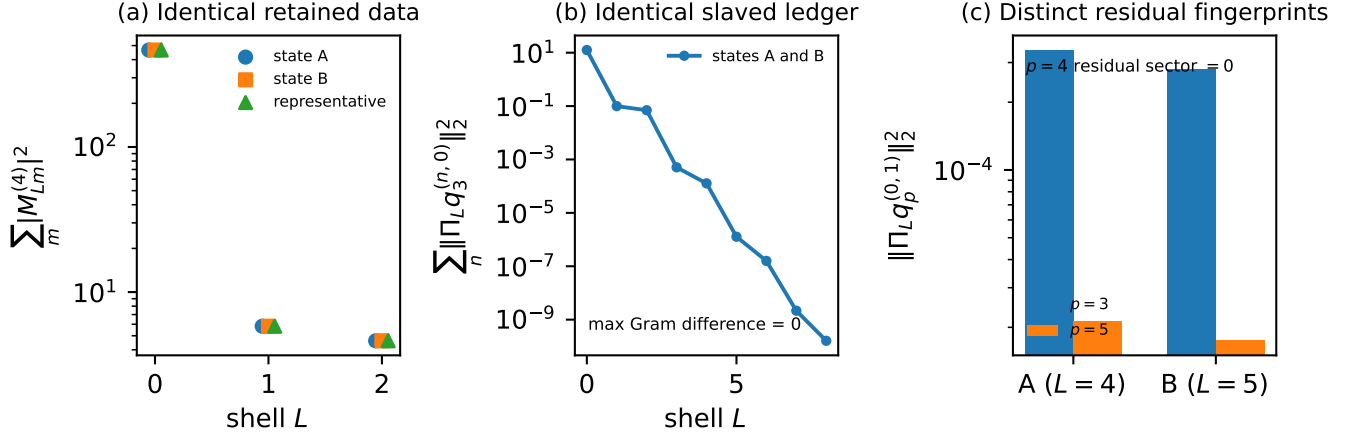


FIG. 5. Equal retained information does not determine residual-dependent observables. (a) States A and B and their common representative have identical $p = 4$ moment shell powers through the retained cutoff. (b) Their representative-generated order-diagonal slaved ledger is identical. (c) The first residual insertion distinguishes the states: A begins in $L = 4$, B in $L = 5$, while the $p = 4$ residual-insertion sector vanishes. The $p = 3$ and $p = 5$ bars differ in magnitude because $c_3 = -1/16$ and $c_5 = 1/64$.

so number-weighted and higher-radial-weight observables respond in opposite directions. For $s = 0.30g_A$ or $0.30g_B$ with the bounded residual harmonics used above, direct radial quadrature verifies pointwise $p = 4$ equality to 7.8×10^{-16} , while the maximum pointwise deviations reach 5.35% for $p = 3$ and 8.25% for $p = 5$. This is a static physical anchor, not a transport benchmark: it demonstrates on a standard temperature-mixture distribution which information the energy-matched representative cannot identify.

IX. NUMERICAL VERIFICATION

Table II collects the principal falsification tests. The thresholds were fixed before the final two-seed ledger runs; they are verification tolerances rather than estimates of physical modeling error. Appendix D gives the quadratures, safeguards, independent-grid checks, and regression criteria.

TABLE II. Summary of the principal numerical checks. “Error“ denotes a relative error unless an absolute quantity is

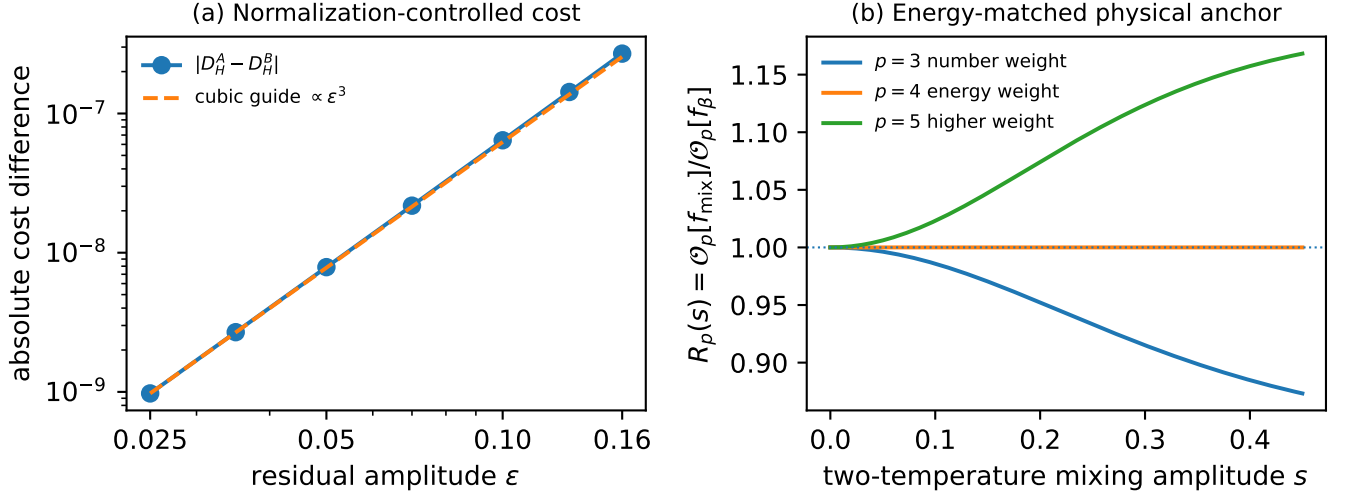


FIG. 6. Normalization control and a physical temperature-mixture anchor. (a) After equalizing the leading Fisher norm, the residual information-cost difference follows the cubic scaling predicted by Eq. (87). (b) For the positive two-temperature mixture in Eq. (89), the $p = 4$ energy moment is exactly preserved while the $p = 3$ number-weighted and $p = 5$ higher-radial-weight moments move in opposite directions.

explicitly identified. The boundary-conditioning row is diagnostic and has no pass threshold.			
Check	Independent setup	Largest observed discrepancy or range	Criterion
Dual derivatives	Finite differences of objective and gradient	1.65×10^{-10} (gradient); 3.15×10^{-11} (Hessian)	10^{-8} ; 10^{-7}
Quantum-statistics lift	Independent MB/BE/FD radial quadratures and nonaxisymmetric recovery	2.74×10^{-16} (constants); 2.52×10^{-15} (recovery); 3.04×10^{-14} (gap)	10^{-10} ; 10^{-10} ; 10^{-9}
Entropy identities	Non-Teff mixture and nested representatives	2.51×10^{-13} (gap); 4.64×10^{-10} (adjacent)	10^{-9}
Matched finite charts	Separate β , u , and Θ solves	Moment residual $\leq 1.04 \times 10^{-13}$; nonnatural deficits positive	10^{-9} residual
Support and parity	Two seeds; $\Lambda = 1, 2, 3$; $n \leq 6$	1.12×10^{-27} above $n\Lambda$; 4.02×10^{-30} wrong parity	10^{-12}
Gram invariance	30 rotations plus 8 on an independent grid	2.04×10^{-15}	10^{-10}
Reconstruction and staircase	Pointwise, harmonic, and discarded-shell checks	1.63×10^{-15} relative; 8.88×10^{-16} absolute	10^{-10}
Response resummation	$p = 3, 4, 5$, $\ v\ _\infty = 0.36$, order 30	Maximum 3.50×10^{-11}	10^{-9}
Local Fisher remainder	Two random seeds	Exponent 3.015–3.082	2.8–3.2
Equal-information control	Independent representative solves	Difference 1.08×10^{-14} ; $p = 4$ residual-insertion term zero	Structural falsifier
Audit normalization	Equal Fisher norm and independent ε sweep	Cost ratio 0.999655; difference exponent 3.028	$ \text{slope} - 3 < 0.2$
Multi-residual and mixture anchor	Two residual coordinates; direct radial mixture quadrature	1.34×10^{-15} reconstruction; 7.8×10^{-16} $p = 4$ error	10^{-11} ; 10^{-12}
Boundary path	min β decreased to 0.005	Condition number up to 9.46×10^5	Diagnostic

A. Dual solver and radial validation

The analytic dual gradient and Hessian were checked against independent finite differences, with maximum relative discrepancies 1.65×10^{-10} and 3.15×10^{-11} , respectively. For the statistics-uniform lift, independent radial

quadratures reproduced the MB, Planck-BE, and zero-chemical-potential FD constants with maximum relative discrepancy 2.74×10^{-16} . Nonaxisymmetric representative recovery and the quantum entropy-gap identity were verified to 2.52×10^{-15} and 3.04×10^{-14} , respectively. The numerical optimizer uses a positivity-preserving damped Newton step; existence in Theorem 3 is analytic and does

not depend on this algorithm.

B. Entropy identities and chart comparison

The independent non-Teff mixture test gives a maximum adjacent-identity discrepancy of 4.64×10^{-10} and shared-moment residual of 3.05×10^{-13} . The separately moment-matched chart comparison satisfies all constraints to 1.04×10^{-13} or better and verifies the strict entropy ordering in Eq. (47); the data and commands underlying Fig. 2 are included in the reproducibility archive.

C. Rotational support, Gram invariance, and cutoff translation

The harmonic-validation suite uses two deterministic random seeds, nonaxisymmetric fields at $\Lambda = 1, 2, 3$, and orders through $n = 6$. Including the independent-grid calculations, the largest relative power above $L = n\Lambda$ is 1.12×10^{-27} , and the largest forbidden-parity fraction is 4.02×10^{-30} . For the insertion ledger through total order four, direct pointwise and harmonic reconstructions agree to at worst 1.64×10^{-15} . The most negative numerical Gram eigenvalue is -1.96×10^{-16} , consistent with roundoff around a positive-semidefinite matrix. Thirty common random rotations on the primary grid and eight on an independent grid change any complete active-shell Gram matrix by at most 2.05×10^{-15} .

The cutoff-translation control keeps an exact nonaxisymmetric V_2 Teff state fixed while changing the retained cutoff. At $\Lambda = 1$ the omitted shell contributes to the residual-dependent and cross sectors, whereas at $\Lambda = 2$ the residual vanishes and all finite-order content is slaved. The total response changes by at most 3.37×10^{-16} relative. For $\|v\|_\infty = 0.36$, the order-30 relative L^2 errors of the resummed $p = 3, 4, 5$ observables are at worst 5.08×10^{-13} , 4.93×10^{-12} , and 3.50×10^{-11} across the two seeds. The local MB Fisher remainder has fitted exponents between 3.015 and 3.082, in agreement with Eq. (71).

The equal-information construction supplies an independent negative control. The retained targets agree numerically, the independently recovered representatives differ by 1.08×10^{-14} , the slaved Gram ledgers agree to machine precision, and the $p = 4$ residual-insertion coefficient is identically zero. These are the checks used in Fig. 5.

D. External-audit controls and physical anchor

The normalization audit reproduces the original L^∞ -normalized information-cost ratio 1.222027 and the corresponding Fisher-norm ratio 1.222025. Equal unweighted L^2 normalization gives cost ratio 0.999492, while equal leading Fisher normalization gives 0.999655. The latter

difference has fitted exponent 3.0275 under an independent amplitude sweep. The finite two-residual expansion in Eq. (58), truncated at total degree 18 for the test amplitudes, reconstructs the direct observable with relative L^2 error 1.34×10^{-15} . For the two-temperature mixture, direct radial quadrature agrees with Eq. (91) to 4.08×10^{-14} and independently recovered representatives differ by 8.96×10^{-15} .

E. Boundary conditioning

On the dipolar path $\beta = 1 + (1 - \delta) \cos \theta$, the solver converges for every tested interior point down to $\min \beta = \delta = 0.005$. The dual-Hessian condition number grows from 2.17×10^1 at $\delta = 0.5$ to 9.46×10^5 at $\delta = 0.005$, with fitted near-boundary scaling approximately $\delta^{-2.09}$. Thus the inverse-cubic barrier secures interior existence but does not imply uniform numerical conditioning near the realizability boundary; this deterioration is directly relevant when strongly anisotropic states drive the minimum directional inverse temperature toward zero.

F. Higher-weight existence versus Teff-class preservation

For the MB family $e^{-\beta E - \gamma E^2}$, a two-parameter dual solve recovers the manufactured target $(\beta, \gamma) = (1.25, 0.075)$ with relative moment error 3.08×10^{-16} and positive Hessian minimum eigenvalue 0.3700. The state is therefore an ordinary higher-weight maximum-entropy representative, while $[-\ln f(E)]/E = \beta + \gamma E$ leaves the linear-tail class. A degenerate target in the same basis returns $\gamma = 0$ and recovers a Teff state. The measured crossover obeys $E_{\text{cross}} \propto \gamma^{-1}$ with log-log slope -0.9999955 . These controls support the class-preservation statement of Theorem 6 and falsify a stronger nonexistence reading.

X. DISCUSSION

The maximum-entropy construction selects a representative; it does not recover information absent from the retained moments. The amplitude ledger makes that distinction observable-specific by retaining the provenance label (n, k) before shell contraction: n records composition order in the representative amplitude, and k records whether a residual not fixed by the moment vector was inserted. An ordinary harmonic spectrum records only the summed output shell and therefore cannot answer this attribution question.

The resulting contract is static but operational. An algebraically slaved shell can be evaluated from the representative alone, but its presence is not evidence that omitted kinetic information has been recovered. A nonzero residual-dependent term identifies the observable sector

that an enriched model would have to supply, but the ledger neither estimates that residual nor proves that an enrichment improves a time-dependent transport calculation. The two-temperature example in Sec. VIII A grounds this distinction in a standard positive kinetic distribution while deliberately stopping short of a transport-performance claim. Such a prediction requires independent kinetic dynamics and reference solutions.

The closest neighboring methods solve different problems. Entropy and ϕ -divergence moment methods construct a closure distribution or a closed moment system from prescribed moments [4, 7, 13]; the present ledger is evaluated after a representative has been selected and does not close the dynamics. Temperature-mixture and temperature-moment formalisms organize spectral mixing and distortion information [14, 15]; here the primary reduction is angular and the insertion count is relative to an angular energy-moment representative. Standard recoupling theory supplies the support algebra [16, 17], while the provenance semantics comes from retaining the representative/residual split before contraction. Finally, Gramian closures use moment Gram matrices to predict a next primitive moment [20]; our shell Gram matrices instead remove orientation within an irreducible output shell while retaining interference between response orders and insertion sectors. Fisher–Rao geometry on a relativistic moment manifold has also been used to quantify geodesic distance, curvature, and hydrodynamic approach [19]; the present Fisher structure is not a dynamical distance construction but the local metric bridge for a representative-conditioned observable ledger. Dynamical projection and invariant-manifold methods address memory, slow manifolds, and transport invariance [22, 23]; by contrast, the present use of *slaved* is instantaneous, representative-relative, and observable-specific. A recent BGK preprint gives a complementary observability warning: accurate low-order macroscopic fields need not certify a tail-weighted fourth-order quantity [24]. A concurrent normal-shock study sharpens that issue into a rank-deficient heat-flux observation map for two fourth-order closure variables and restores identifiability with a second scalar budget [25]. A related rarefied-cavity study finds a larger functional null space, including divergence-free tensor fields and an exactly invisible out-of-plane channel, so agreement of the anti-Fourier heat flux does not certify the complete fourth-order closure state [26]. The present result addresses a different map: a static angular energy-moment representative with a pointwise $p = 4$ null residual. It proves nonidentifiability and records residual-dependent observable provenance, but it does not propose the additional dynamical recovery channels considered in those flow problems.

Three exact structures must remain separate. The entropy gap and adjacent-resolution identity are consequences of moment matching in distribution space. The Gram staircase is an exact L^2 identity for an angular observable. Their connection is local and Fisher weighted, becoming shell diagonal at an isotropic base state but not

generally away from isotropy. Combining them without this firewall would turn a useful quadratic bridge into a false global entropy equality.

The finite-coordinate comparison adds a second caution. The full positive fields β , $u = -\ln(\beta/\beta_0)$, and $\Theta = \beta_0/\beta$ are equivalent coordinates, but their finite harmonic submanifolds are not. Even when the u and Θ competitors are fitted to the same complete retained moment vector, the natural inverse-temperature representative has the largest entropy. A finite “temperature truncation” is therefore part of the model specification rather than a coordinate-neutral operation.

The present results concern the massless, one-field, admissible interior. They do not establish a boundary representative, uniform conditioning near realizability loss, a global two-field $\{0,1\}$ diffeomorphism, a Bose-condensate sector, a transport closure, or global-in-time stability. The class-relative minimality result likewise does not deny the existence or utility of higher-order entropy closures; it states only that independent higher energy multipliers leave the finite-positive linear Wien-tail class on a full-dimensional parameter domain.

For transport applications, the natural next step is dynamical rather than algebraic: evolve a representative, monitor a collision- or streaming-induced residual, and test whether insertion-resolved provenance predicts observable error before a spectral or angular enrichment is activated. That step requires independent Boltzmann or multigroup reference solutions and is intentionally outside the present foundational article.

XI. CONCLUSION

Every interior realizable angular energy-moment vector selects a unique maximum-entropy inverse-temperature representative for the one-field MB, Planck-BE, and zero-chemical-potential FD sectors. Nested moment sets produce exact entropy inequalities, while analytic observables admit a triangular amplitude-shell ledger augmented by a residual-insertion count. That coordinate-relative count separates high-multipole content fixed by retained data from content requiring additional kinetic information relative to the chosen cutoff, and $SO(3)$ -invariant Gram matrices provide an exact shellwise and discarded-block account. The equal-information counterfamily establishes the sharp limit: maximum-entropy reduction determines the representative and every algebraically slaved nonlinear consequence of it, but it cannot determine residual-dependent observables of the exact state.

Appendix A: Convex conjugates, Planck-edge bounds, and the interior representative theorem

1. Conjugates and the stationary family

The conjugate is

$$h_\xi^*(y) = \sup_f \{yf - h_\xi(f)\}. \quad (\text{A1})$$

The stationarity equation is

$$y = h'_\xi(f) = \ln \frac{f}{1 + \xi f}, \quad (\text{A2})$$

so

$$f = \frac{e^y}{1 - \xi e^y} = \frac{1}{e^{-y} - \xi}. \quad (\text{A3})$$

Substitution gives Eq. (19). The Lagrangian for Eq. (12) is

$$\mathcal{L}[f, b] = H_\xi[f] + \sum_i b_i (M_i[f] - m_i). \quad (\text{A4})$$

At an interior stationary point,

$$h'_\xi(f) + x\beta_b = 0, \quad (\text{A5})$$

which, by Eq. (A3), gives Eq. (13). Thus every interior stationary point belongs to the inverse-temperature family.

2. Radial constants and the Planck edge

For MB,

$$\int_0^\infty s^2 e^{-s} ds = 2. \quad (\text{A6})$$

For BE, the nonnegative expansion

$$-\ln(1 - e^{-s}) = \sum_{k=1}^\infty \frac{e^{-ks}}{k} \quad (\text{A7})$$

gives, by monotone convergence,

$$\int_0^\infty s^2 [-\ln(1 - e^{-s})] ds = \sum_{k=1}^\infty \frac{2}{k^4} = 2\zeta(4). \quad (\text{A8})$$

For FD,

$$\ln(1 + e^{-s}) = \sum_{k=1}^\infty \frac{(-1)^{k+1} e^{-ks}}{k}; \quad (\text{A9})$$

termwise integration is dominated by the integrable BE series and yields

$$\int_0^\infty s^2 \ln(1 + e^{-s}) ds = 2 \sum_{k=1}^\infty \frac{(-1)^{k+1}}{k^4} = 2(1 - 2^{-3})\zeta(4). \quad (\text{A10})$$

This proves Eqs. (22) and (23).

For BE, as $s \rightarrow 0^+$,

$$h_{+1}^*(-s) = -\ln(1 - e^{-s}) = O(|\ln s|), \quad (\text{A11})$$

whereas

$$f_{+1}(s) = O(s^{-1}), \quad f_{+1}(s)[1 + f_{+1}(s)] = O(s^{-2}). \quad (\text{A12})$$

Consequently, at fixed $\beta > 0$, the radial integrands for the potential, moment, and Hessian behave as

$$O(x^2 |\ln x|), \quad O(x^2), \quad O(x^2), \quad (\text{A13})$$

respectively. On a compact positive subcone $0 < \beta_- \leq \beta_b \leq \beta_+$, the estimates $e^t - 1 \geq t$ for $0 < t \leq 1$ and exponential bounds for $t \geq 1$ supply parameter-independent integrable dominators. This proves Lemma 2. The factorized expression then makes smoothness on $\mathcal{B}_\Lambda$ explicit.

3. Cone geometry and the boundary barrier

Because the harmonic representation is injective, K_Λ is pointed. The positive constant mode makes it full dimensional and ensures $\mathcal{B}_\Lambda \neq \emptyset$. For $b \in \mathcal{B}_\Lambda$ and nonzero $c \in K_\Lambda$,

$$c \cdot M_\xi(b) = 3a_\xi C \int_{\mathbb{S}^2} \beta_c \beta_b^{-4} d\Omega > 0, \quad (\text{A14})$$

so $M_\xi(b) \in \text{int } \mathcal{R}_\Lambda$.

Lemma 23 (Inverse-cubic boundary barrier). *If a bounded sequence $b_n \in \mathcal{B}_\Lambda$ approaches $\partial\mathcal{B}_\Lambda$, then $A_{\xi,\Lambda}(b_n) \rightarrow +\infty$.*

Proof. Let $\varepsilon_n = \min_{\mathbb{S}^2} \beta_{b_n} \rightarrow 0$ and choose a minimizer $\hat{e}_n$. Finite dimensionality and bounded coefficients give a finite uniform spherical-gradient bound. Choose any positive constant L strictly larger than that bound. On the geodesic ball of radius $r_n = \varepsilon_n/L$ around $\hat{e}_n$,

$$\beta_{b_n} \leq 2\varepsilon_n. \quad (\text{A15})$$

For sufficiently small r_n , the ball area is at least $c_0 r_n^2$. Hence

$$A_{\xi,\Lambda}(b_n) \geq a_\xi C c_0 r_n^2 (2\varepsilon_n)^{-3} = \frac{a_\xi C c_0}{8L^2} \varepsilon_n^{-1} \rightarrow \infty. \quad (\text{A16})$$

□

4. Coercivity, surjectivity, and smooth inversion

Fix $m \in \text{int } \mathcal{R}_\Lambda$. Compactness of $K_\Lambda \cap S^{N_\Lambda-1}$ gives

$$\delta_m = \min_{c \in K_\Lambda, \|c\|=1} c \cdot m > 0. \quad (\text{A17})$$

Therefore, for $b \in \mathcal{B}_\Lambda$,

$$b \cdot m \geq \delta_m \|b\|. \quad (\text{A18})$$

Equations (29), (A18), and Lemma 23 show that every sublevel set of $\Psi_{\xi,m}$ is compactly contained in $\mathcal{B}_\Lambda$. The function attains a unique interior minimum because it is strictly convex. Its critical equation is

$$\nabla \Psi_{\xi,m} = m - M_\xi(b) = 0, \quad (\text{A19})$$

which proves surjectivity of Eq. (27). Strict convexity makes the gradient $\nabla A_{\xi,\Lambda}$ injective, hence so is $M_\xi = -\nabla A_{\xi,\Lambda}$. Equation (26) and the inverse-function theorem then give a smooth inverse.

Finally, compactness of $\mathbb{S}^2$ and $\beta_b > 0$ give finite positive lower and upper bounds on β_b . Together with the MB/FD exponential tails and the Planck-edge low-energy estimate above, these bounds ensure that the constructed state has finite retained moments and finite entropy and is feasible for the primal problem. Let $f_{\xi,m}^*$ denote this state and let f be any other feasible state. Since

$$h'_\xi(f_{\xi,m}^*) = -x\beta_b = -x \sum_i b_i Z_i, \quad (\text{A20})$$

moment matching implies

$$\int h'_\xi(f_{\xi,m}^*)(f - f_{\xi,m}^*) d\Pi = 0. \quad (\text{A21})$$

Equation (8) therefore reduces to

$$H_\xi[f] - H_\xi[f_{\xi,m}^*] = D_{H_\xi}(f \| f_{\xi,m}^*) \geq 0, \quad (\text{A22})$$

with equality only almost everywhere. This completes the proof of Theorem 3.

5. Gram-Hessian calculation

The tangent vector associated with δb is

$$\delta f = -x f_{\xi,b}(1 + \xi f_{\xi,b}) \delta \beta. \quad (\text{A23})$$

Using Eq. (6),

$$\delta^2 H_\xi[f_{\xi,b}](\delta f, \delta f) = \int \frac{(\delta f)^2}{f_{\xi,b}(1 + \xi f_{\xi,b})} d\Pi \quad (\text{A24})$$

$$= \int x^2 f_{\xi,b}(1 + \xi f_{\xi,b})(\delta \beta)^2 d\Pi \quad (\text{A25})$$

$$= \delta b^T \nabla^2 A_{\xi,\Lambda}(b) \delta b. \quad (\text{A26})$$

Multiplication by $-k_B$ gives Eq. (31).

Appendix B: Polynomial-tail classification and codimension proof

1. Directionwise polynomial lemma

Let

$$p(E) = \sum_{m=0}^d a_m E^m. \quad (\text{B1})$$

Then

$$0 < \lim_{E \rightarrow \infty} \frac{p(E)}{E} < \infty \quad (\text{B2})$$

if and only if

$$a_m = 0 \quad (m \geq 2), \quad a_1 > 0. \quad (\text{B3})$$

Indeed,

$$\frac{p(E)}{E} = \frac{a_0}{E} + a_1 + \sum_{m=2}^d a_m E^{m-1}. \quad (\text{B4})$$

If $d_* \geq 2$ is the highest degree with $a_{d_*} \neq 0$, the last expression is $a_{d_*} E^{d_*-1}[1 + o(1)]$ and diverges with the sign of a_{d_*} . Conversely, Eq. (B3) gives the finite positive limit a_1 .

2. Angular lifting and empty interior

Applying the directionwise lemma to Eq. (32) proves that Eq. (33) holds if and only if

$$\lambda_m(\hat{e}) = 0 \quad (m \geq 2), \quad \lambda_1(\hat{e}) > 0 \quad (\text{B5})$$

for every $\hat{e}$. If vanishing is first known only almost everywhere, continuity supplies the pointwise result: a continuous coefficient nonzero at one point is nonzero on an open neighborhood of positive measure.

Define

$$\mathcal{K}_\mathcal{W} = \{\lambda \in \mathcal{P}_\mathcal{W} : \lambda_m = 0 \text{ for all } m \geq 2\}. \quad (\text{B6})$$

If a higher degree is present, this is a closed linear subspace with codimension Eq. (36). It therefore has empty interior in $\mathcal{P}_\mathcal{W}$, and the class-preserving subset is contained in

$$\mathcal{K}_\mathcal{W} \cap \{\lambda_1 > 0\}, \quad (\text{B7})$$

with any low-energy restrictions imposed separately. If no higher degree is present, Eq. (B5) defines a nonempty open cone precisely when the linear weight is included. This proves Theorem 6.

3. Countercontrols and finite-window mimicry

For MB statistics, Eq. (38) satisfies

$$h'_0(f_{\beta,\gamma}) = -\beta E - \gamma E^2. \quad (\text{B8})$$

Under the corresponding linear and quadratic moment constraints, the same Bregman cancellation as in Eq. (A22) proves that $f_{\beta,\gamma}$ is the unique entropy minimizer for its own target whenever $\beta, \gamma > 0$. At $\gamma = 0$, the larger family recovers the Teff member exactly. Thus the presence of a higher basis function neither implies nonexistence nor excludes a lower-dimensional Teff solution.

For $0 < \gamma \ll 1$, the quadratic correction becomes comparable to the linear term near

$$E_{\text{cross}} \sim \frac{\beta}{\gamma}. \quad (\text{B9})$$

A finite energy window $E \ll E_{\text{cross}}$ may therefore resemble a linear-tail state even though the asymptotic class differs. Numerical tail sweeps in Sec. IX are used only as falsification controls for this distinction; they are not substitutes for the polynomial proof.

Appendix C: Angular recoupling, convergence, and Gram proofs

1. Harmonic conventions and Gaunt coefficients

For complex fields the shell inner product is Hermitian,

$$\langle a, b \rangle = \int_{\mathbb{S}^2} a(\hat{e})^* b(\hat{e}) d\Omega, \quad (\text{C1})$$

with the real-basis formula obtained by dropping the conjugation.

Let Y_{Lm} be the standard complex orthonormal harmonics,

$$\int_{\mathbb{S}^2} Y_{Lm}^* Y_{L'm'} d\Omega = \delta_{LL'} \delta_{mm'}, \quad Y_{L,-m} = (-1)^m Y_{Lm}^*. \quad (\text{C2})$$

Any real orthonormal basis Z_{La} is related to the complex basis by a unitary transformation restricted to the L shell. Therefore support statements and shell Gram matrices are basis independent.

The product formula is

$$Y_{\ell_1 m_1} Y_{\ell_2 m_2} = \sum_{LM} \mathcal{G}_{\ell_1 m_1, \ell_2 m_2}^{LM} Y_{LM}, \quad (\text{C3})$$

where

$$\begin{aligned} \mathcal{G}_{\ell_1 m_1, \ell_2 m_2}^{LM} &= (-1)^M \left[\frac{(2\ell_1 + 1)(2\ell_2 + 1)(2L + 1)}{4\pi} \right]^{1/2} \\ &\times \begin{pmatrix} \ell_1 & \ell_2 & L \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell_1 & \ell_2 & L \\ m_1 & m_2 & -M \end{pmatrix}. \end{aligned} \quad (\text{C4})$$

The $3j$ symbols vanish unless

$$|\ell_1 - \ell_2| \leq L \leq \ell_1 + \ell_2, \quad M = m_1 + m_2, \quad \ell_1 + \ell_2 + L \text{ even}. \quad (\text{C5})$$

These are the standard angular-momentum addition rules [16].

2. Proof of hard support and parity

Expand each factor in v_R^n into harmonics of rank at most Λ . Applying Eq. (C5) to the first two factors gives

an intermediate rank no larger than their sum. Coupling the next factor increases the largest possible rank by at most its rank. Induction yields

$$L \leq \ell_1 + \dots + \ell_n \leq n\Lambda, \quad (\text{C6})$$

which proves Eq. (53). Under inversion, $Y_{Lm}(-\hat{e}) = (-1)^L Y_{Lm}(\hat{e})$. If v_R has parity $(-1)^\sigma$, then v_R^n has parity $(-1)^{n\sigma}$, so only shells with $L \equiv n\sigma \pmod{2}$ survive. Mixed-parity inputs have no single global parity restriction beyond the termwise Gaunt rules.

3. Rotation invariance and basis independence

Let $U(R)f(\hat{e}) = f(R^{-1}\hat{e})$. The L shell is invariant under $U(R)$, and the restriction $U_L(R)$ is unitary. Therefore

$$\Pi_L U(R) = U(R) \Pi_L, \quad (\text{C7})$$

and for every pair of response labels,

$$G_L^{ab}[U(R)q] = \langle U_L(R)s_L^a, U_L(R)s_L^b \rangle = G_L^{ab}[q]. \quad (\text{C8})$$

An orthonormal basis change inside the same shell is another unitary map and gives the same conclusion. This proves the invariance part of Theorem 18; positivity was proved in Eq. (63).

4. Proof of the exact staircase

Because Q_Λ is linear and $\mathcal{J} = \sum_a q^a$ in L^2 ,

$$\mathbf{1}^\dagger G_{>\Lambda} \mathbf{1} = \sum_{a,b} \langle Q_\Lambda q^a, Q_\Lambda q^b \rangle \quad (\text{C9})$$

$$= \left\| Q_\Lambda \sum_a q^a \right\|_2^2. \quad (\text{C10})$$

Furthermore,

$$Q_\Lambda = \Pi_{\Lambda+1} + Q_{\Lambda+1}, \quad \Pi_{\Lambda+1} Q_{\Lambda+1} = 0, \quad (\text{C11})$$

so Pythagoras gives Eq. (67).

5. Uniform binomial remainder

For an integer $p \geq 1$ and $\rho = \|v\|_\infty < 1$,

$$\frac{(p)_n}{n!} = \binom{p+n-1}{p-1}. \quad (\text{C12})$$

Hence the remainder after order N obeys

$$\begin{aligned} \left\| (1+v)^{-p} - \sum_{n=0}^N c_n^{(p)} v^n \right\|_\infty &\leq \sum_{n=N+1}^\infty \binom{p+n-1}{p-1} \rho^n \\ &\leq C_p(\rho) (N+p)^{p-1} \rho^{N+1}, \end{aligned} \quad (\text{C13})$$

where

$$C_p(\rho) = \frac{1}{(p-1)!} \sum_{j=0}^{\infty} (j+1)^{p-1} \rho^j < \infty. \quad (\text{C14})$$

The second inequality uses $\binom{p+N+j}{p-1} \leq (N+p)^{p-1} (j+1)^{p-1} / (p-1)!$. Multiplication by $\sqrt{4\pi}$ gives the corresponding L^2 bound on $\mathbb{S}^2$.

Appendix D: Numerical algorithms, safeguards, and regression criteria

1. Angular quadrature and the dual solve

The inverse-temperature field is expanded in a real orthonormal harmonic basis. Angular integrals for the representative problem use a tensor product of Gauss–Legendre quadrature in $\mu = \cos \theta$ and a uniform azimuthal rule. Optimization and validation use different node counts and nonzero relative azimuthal offsets. Across the representative, entropy, and chart tests, optimization grids range from 57×117 to 84×169 nodes, while independent validation grids range from 121×241 to 139×277 nodes. The basis orthonormality matrix is checked on every grid.

For a target moment vector m , the finite-dimensional objective is

$$\Psi_m(b) = A_{\xi, \Lambda}(b) + b \cdot m, \quad \nabla \Psi_m = m - M_{\xi}(b), \quad (\text{D1})$$

with Hessian given by Eq. (25). A Cholesky factorization solves

$$\nabla^2 \Psi_m(b_j) \Delta b_j = -\nabla \Psi_m(b_j). \quad (\text{D2})$$

The initial state is the isotropic field matching the energy monopole. Before backtracking, the step length is capped at 0.995 of the first grid point at which $\beta_b + \alpha \Delta \beta$ would vanish. Backtracking halves α until either the Armijo condition with coefficient 10^{-4} or a corresponding residual-decrease condition is met. The standard termination test is

$$\|\nabla \Psi_m\|_2 \leq 2 \times 10^{-12} \max(1, \|m\|_2), \quad (\text{D3})$$

with at most 120 iterations; the dedicated boundary sweep permits 160 iterations. The theorem of Sec. III is independent of this algorithm.

2. Radial integration and positivity certification

The analytic radial factorization is checked in two independent ways. One implementation uses generalized Gauss–Laguerre quadrature with weight $x^2 e^{-x}$ and order 96 or higher. A second implementation maps Gauss–Legendre nodes to $(0, \infty)$ and uses 210–230 radial nodes in the entropy-gap and adjacent-resolution tests. The

exact-state test is a positive two-temperature mixture, so agreement is not an inverse check on a manufactured one-field state.

Positivity on the optimization nodes is enforced by the line search. A separate sufficient certificate controls the continuous sphere. If $b_{\ell a}$ are the harmonic coefficients, the addition theorem and Cauchy–Schwarz give

$$\|\nabla_{\mathbb{S}^2} \beta\|_{\infty} \leq \left(\sum_{\ell \geq 1, a} |b_{\ell a}|^2 \right)^{1/2} \left[\sum_{\ell=1}^{\Lambda} \frac{\ell(\ell+1)(2\ell+1)}{4\pi} \right]^{1/2}. \quad (\text{D4})$$

For an equiangular net with covering-radius upper bound ρ ,

$$\min_{\mathbb{S}^2} \beta \geq \min_{\text{grid}} \beta - \rho \|\nabla_{\mathbb{S}^2} \beta\|_{\infty}. \quad (\text{D5})$$

Standard-interior states were checked on nets of at least 141×280 points; the reusable implementation defaults to 361×720 . A positive lower bound certifies positivity. Failure of this sufficient test is only inconclusive and is not interpreted as evidence of a negative field.

3. Harmonic ledgers, rotations, and independent grids

The ledger calculations use complex spherical harmonics for projection and impose the reality condition $a_{\ell, -m} = (-1)^m a_{\ell m}^*$. The primary grid has 32 Gauss–Legendre nodes, 64 azimuthal nodes, and harmonic output through $L = 20$. An independent 36×72 grid recomputes the central Gram family. Random real input fields are normalized by their sampled L^{∞} amplitude. The support suite uses five fields at each $\Lambda = 1, 2, 3$ and response orders through $n = 6$. Rotations are sampled from the Haar distribution on $SO(3)$; 30 common rotations are applied on the primary grid and 8 additional rotations on the independent grid. Two deterministic seeds, 20260624 and 20260625, are retained in the archive.

For generated columns q^a , harmonic projection is performed before shell contraction. The numerical Gram matrix is symmetrized before its eigenvalues are inspected. Direct pointwise reconstruction, harmonic reconstruction, the shellwise slaved/residual/cross identity, cutoff translation, and the exact discarded-content staircase are evaluated independently rather than inferred from the Gram construction.

4. Acceptance criteria and reproducibility

The regression gates are: forbidden support and parity fractions below 10^{-12} ; Gram rotation, cross-grid, reconstruction, staircase, and cutoff-translation discrepancies below 10^{-10} ; the smallest numerical Gram eigenvalue above -10^{-12} ; order-30 response error below 10^{-9} for

$p = 3, 4, 5$; and a fitted Fisher-remainder exponent between 2.8 and 3.2. The derivative and entropy tests use the separate thresholds listed in Table II. All figures are generated from saved comma-separated or JSON data using deterministic scripts; the archive also contains the exact commands, dependency versions, and regression logs.

Appendix E: Claim status and explicit nonclaims

Table III records the logical status of the load-bearing statements. “Exact” always means exact under the hypotheses in the corresponding statement, not unrestricted validity on the boundary or under transport evolution.

TABLE III. Claim-status ledger. The final rows are explicit exclusions, included to prevent the local or class-relative results from being read as stronger global statements.

Statement	Status	Scope and hypotheses	Location
Interior representative and diffeomorphism	Exact	Massless one-field MB, Planck-BE, and zero- μ FD; finite cutoff; $m \in \text{int } \mathcal{R}_\Lambda$	Thm. 3
Entropy gap, filtration, adjacent identity	Exact	Interior moment-matched representatives at nested cutoffs	Sec. V
Natural-chart entropy dominance	Exact	Comparison state matches the same complete retained vector	Prop. 10
Linear-tail class	Exact,	Finite monomial exponent families and a full-dimensional	Thm. 6
preservation	class-relative	open parameter set	
Hard support and parity	Exact	Fixed polynomial response order and bandlimited representative	Thm. 12
Residual-insertion split	Exact	Observable analytic in the chosen bookkeeping amplitudes; fixed response/residual coordinates and scaling path	Def. 13
Finite multi-residual ledger	algebraic	Finite chosen residual coordinates; infinite-dimensional completeness and a canonical basis are not claimed	Rem. 15
$SO(3)$ Gram positivity and invariance	Exact	Generated fields in $L^2(\mathbb{S}^2)$; common rotation and within-shell orthonormal basis changes	Thm. 18
Discarded-content staircase	Exact	Convergent response sum in $L^2(\mathbb{S}^2)$	Thm. 19
Response resummation	Conditional	$\ v\ _\infty < 1$; finite-order polynomial identities do not require resummation	App. C
Fisher–Bregman bridge	Local	Admissible segment and integrable third-order remainder; shell diagonal only at isotropy	Eq. (71)
Energy-matched temperature mixture	Exact static example	Positive MB two-temperature mixture with $ s < 1$; not a transport benchmark	Sec. VIII A
Boundary representative or uniform conditioning	Not claimed	No theorem on $\partial\mathcal{R}_\Lambda$; conditioning deteriorates as $\min \beta \rightarrow 0^+$	Sec. X
Global two-field $\{0, 1\}$ diffeomorphism	Not claimed	Two-field family used only in the class-preservation observation	Sec. X
Transport closure or global stability	Not claimed	No time-dependent solver, microphysical collision benchmark, or global error propagation	Sec. X
Global entropy identity for the Gram staircase	Not claimed	Staircase exact in observable L^2 ; entropy connection local and Fisher weighted	Secs. VII, X
Coordinate-free (n, k) provenance	Not claimed	Nonlinear response-coordinate or residual-coordinate changes can redistribute ledger cells	Rem. 14

DATA AVAILABILITY

A versioned reproducibility archive is supplied with this manuscript. It contains the numerical data, deterministic seeds, executable validation campaigns, regression thresholds, figure-generation commands, environment record, and checksums needed to reproduce the reported results. No proprietary or externally restricted data were used. The same archive will be deposited in a persistent public repository before final submission.

AI USE DISCLOSURE

OpenAI GPT-5.5 Pro (accessed June 2026) was used through author-specified prompts for literature organization, code drafting and review, adversarial claim checking, and language and structure revision. Its outputs were independently checked against primary sources, analytic derivations, and deterministic regression tests; no research data or manuscript figures were generated from unverified model output. It was not listed as an author. The human authors retain full responsibility for the derivations, source selection, numerical results, interpretation, and final manuscript.

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